

Fetal Health Classification from Cardiotocogram Data

"Multiclass Classification of Fetal Health using Cardiotocogram Data" (Ereyi & Huo, Stanford CS229, 2021)

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Problem and dataset

Problem

Classify each CTG record as Normal, Suspect or Pathological from numeric features. Manual reading is slow and subjective.

Data

UCI Cardiotocography: 2,126 records, 21 features, labelled by expert obstetricians. 2,113 used after preprocessing, reduced to 10 features and standardised.

Class	Train	Test
Normal	1,149	497
Suspect	202	90
Pathological	128	47
Total	1,479	634

70/30 train-test split. About 78% of records are Normal.

Approach and implementation

Split data 70/30

**SelectKBest: 10
features,
standardise**

**Train and tune
classifiers**

**Compare with the
paper**

Notebook 03: MLP, Gradient Boosting, XGBoost, LightGBM, Linear SVM

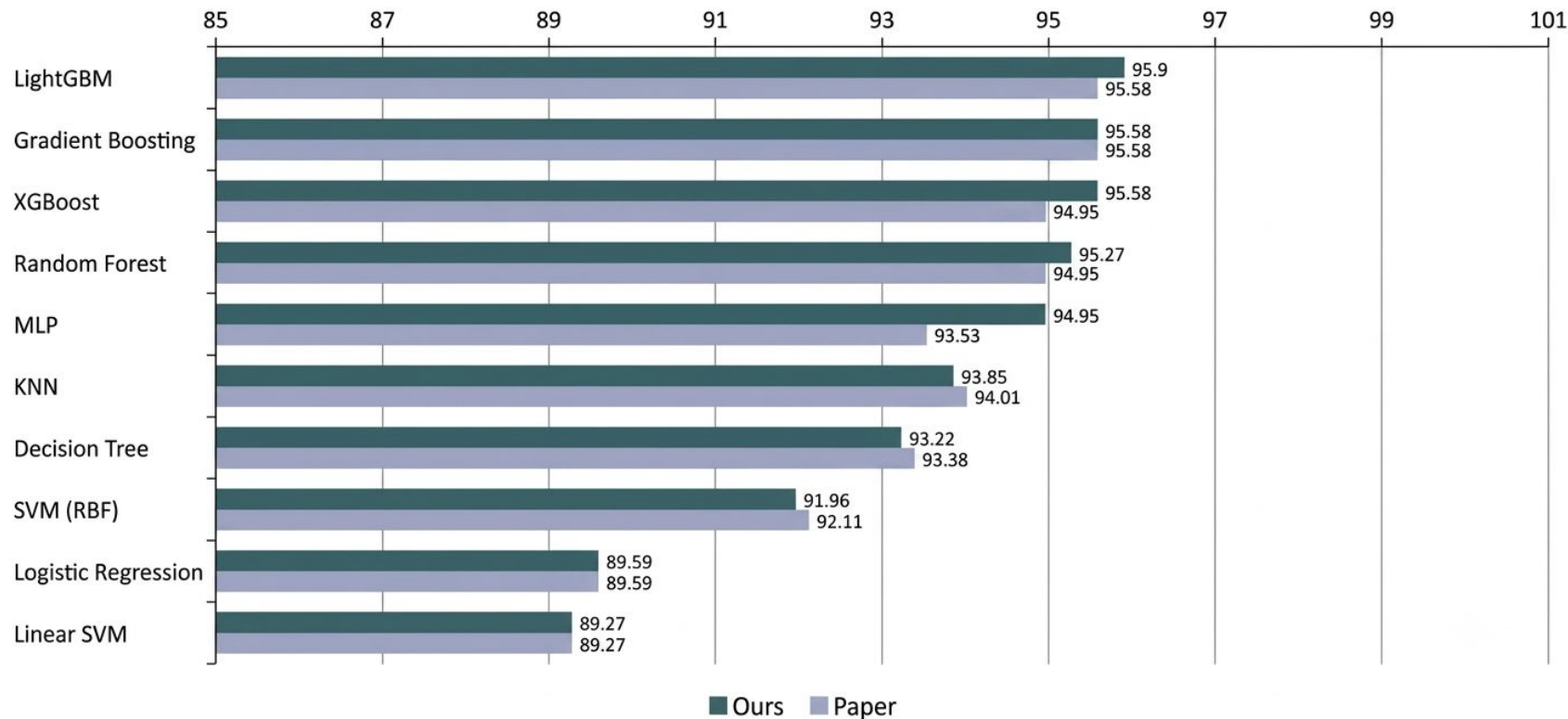
Notebook 04: Logistic Regression, KNN, Decision Tree, Random Forest, SVM

Experiments: grid search, class weights, SMOTE, voting, stacking

Tools: Python, scikit-learn, XGBoost, LightGBM, Google Colab

Metrics: accuracy, per-class precision, recall and F1, confusion matrices

Accuracy: ours vs the paper



Best model from the paper: LightGBM

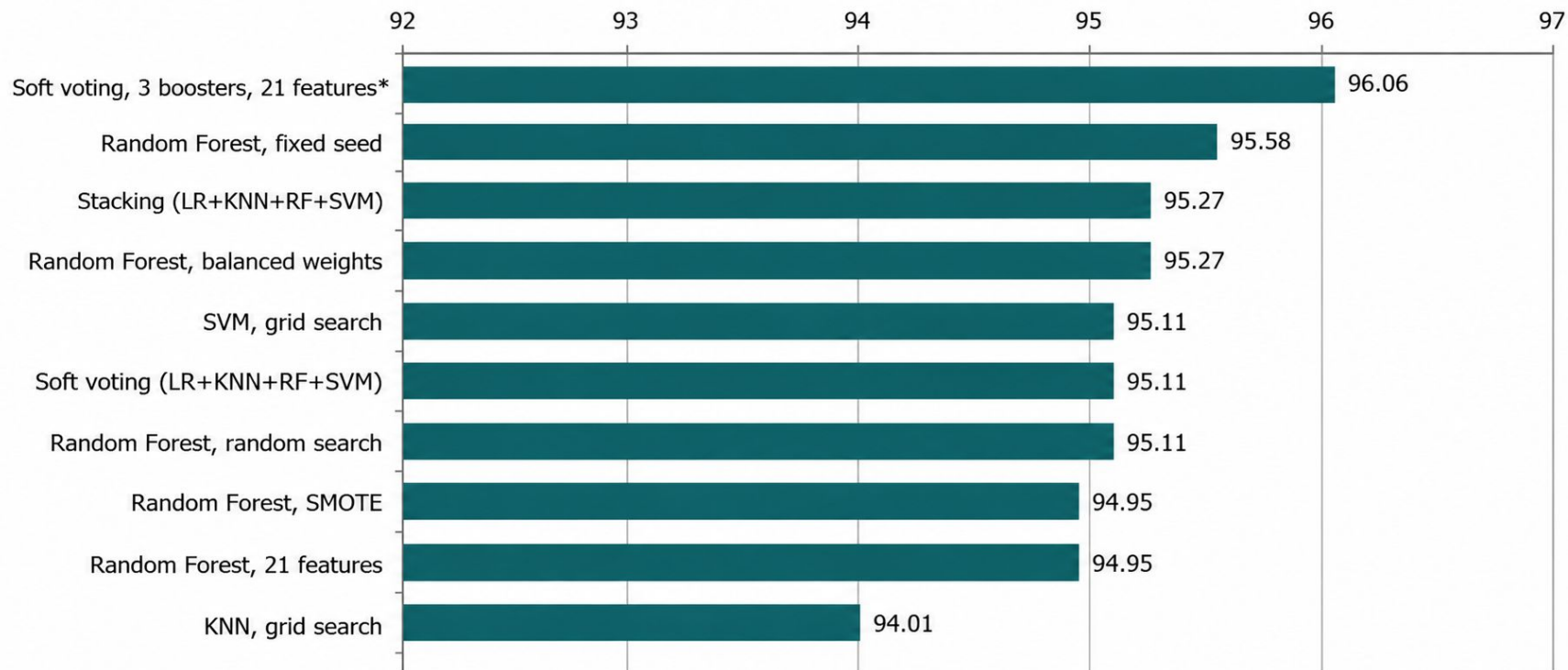
95.90%

test accuracy, 608 of 634 correct (paper: 95.58%)

Class	Precision	Recall	F1
Normal	0.967	0.990	0.978
Suspect	0.947	0.789	0.861
Pathological	0.900	0.957	0.928

Suspect is the hardest class: it is small and overlaps with Normal.

Trying to improve the baseline



Soft voting ensemble

Part	Original experiments	Soft voting
Features	Top 10 (SelectKBest)	All 21
Scaling	StandardScaler	None
Models	Individual classifiers	LightGBM + XGBoost + CatBoost
Prediction	One model predicts	Average class probabilities, equal weights
Split	Existing processed split	New 70/30 stratified, seed 42
Evaluation	Accuracy, report	Accuracy, balanced accuracy, macro F1

96.06 %

Best result overall

Conclusions

- Boosted trees and Random Forest were best (95.3 to 95.9%). Nine of ten models are within 0.65 points of the paper.
- Tuning helped weak models most (SVM +3.2). Voting, stacking, SMOTE and class weights gave no clear gain on one split.
- Soft voting of LightGBM, XGBoost and CatBoost on 21 features reached 96.06 %, but only a few samples above LightGBM (one sample is 0.16 points).
- Limitations: one split, no cross-validation, and some choices compared on the test set.